Noise Adding

Alex Le

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1 Velocity Aberration

1.1 Intro To Velocity Abberation

Velocity aberration occurs when relative movement between an object and the observer results in a tilted viewing angle because the speed of light is finite.

Intuitively, think of it like this:

Let's say it is raining on a windless day. The rain falls straight down. If you are stationary, you would point your umbrella directly upwards.

But when you begin running forwards, the rain appears to "go towards you" from *your* perspective. So you have to tilt your umbrella forwards. The faster you run, the more you tilt your umbrella forwards.

1.2 Why It Matters

Although satellites are nonrelativistic ($\beta \approx 10^{-5}$), they indeed have to account for velocity aberration both for attitude control and ground pictures.

On-board star trackers can accumulate errors of around $\sim 125\mu\mathbf{rad}$ if they do not account for the fact that they have significant velocity around the Earth, which itself is moving around the Sun [1]

Similarly, satellites looking down at the Earth have to account for aberration effects as well. A 700 km high satellite has an orbital speed of ~ 7 **km**. This causes an angular deviation of $23\mu\mathbf{rad}$, corresponding to a ground distance error of 20 **meters**. This is not negligible for high-resolution imagery [2]

1.3 How Does It Work?

Let us assume that we are working **non**relativistically; $\beta << 1$. This is reasonable because satellites move slowly.

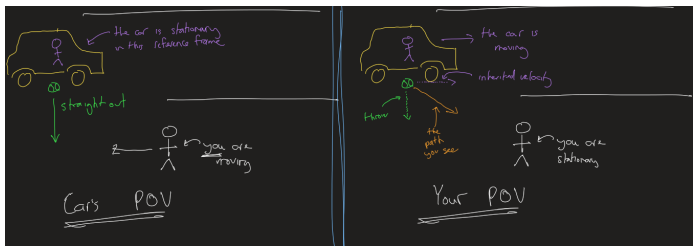
Then, we may use the Galilean Transform instead of the Lorentz Transform, and need not worry about time dilation, length contraction, or other relativistic effects.

In simple words: when a source[1] (i.e. satellites, the ground, or stars) emits light while moving relative to a stationary observer, the light ray appears to

[1] Doesn't matter if they are reflectors or emitters. Source is source.

"inherit" the source's velocity vector. This causes the light's apparent direction to skew a bit.

Imagine that you (an observatory) are standing on the sidewalk and a car (satellite) drives by in front of you, left-to-right. Let's say the car is throwing a tennis ball (photon) at you. From the car's perspective, it throws it directly outwards at a right angle relative to the car's direction of motion. However, from *your* perspective, you must catch the ball as it approaches from your left side, moving towards you *and* right.



2 Application To Satellite Tracking

Because satellites are moving light sources, the light bouncing off of the satellite inherits the satellite's velocity. We'll ease into the problem with progressively harder cases.

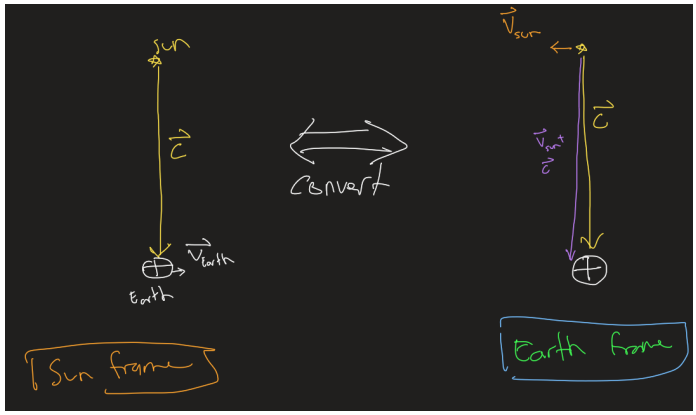
Carefully note: we aren't really interested in the aberration angle. We are more interested in the "apparent location" caused by aberration effects; that is, what the satellite's $\vec{r}$ *appears* to look like

2.1 Simplest Case: Orbit Around The Sun

Assume that the Earth's orbit is perfectly circular.

Define $\vec{c}$ to be the unaberrated trajectory light takes between the sun and the Earth (basically, what would have happened if eveything was stationary). Define $\vec{v}_{\mathbf{Earth}}$ to the velocity vector of the Earth, *once again from the sun's source's point of view.*

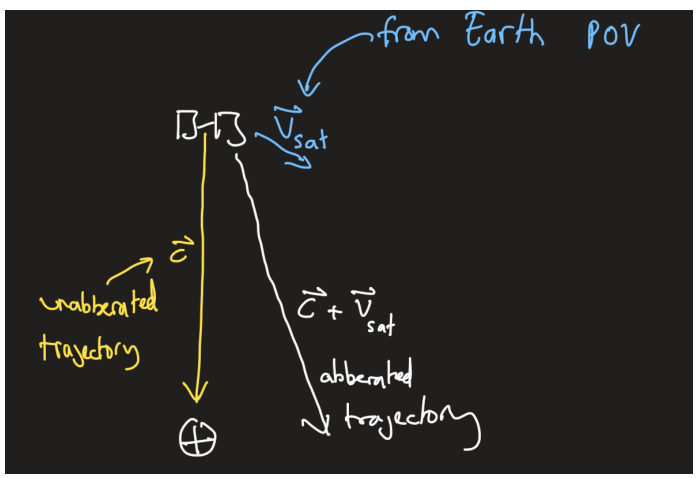
We will change coordinate frames to be from an Earthling's point of view. From our perspective, the Earth is stationary while the sun seems to move backwards; $\vec{v}_{\mathbf{Sun}} = -\vec{v}_{\mathbf{Earth}}$



The path that the light *appears* to take is $\vec{c} - \vec{v}_{\mathbf{Earth}}$ or $\vec{c} + \vec{v}_{\mathbf{Sun}}$

2.2 Slightly Harder Case: Satellite in an elliptical orbit

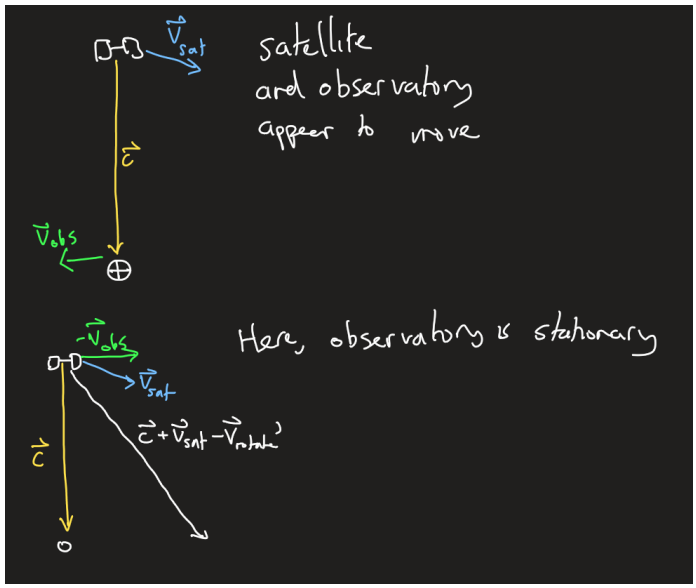
Let's say that the Earth doesn't rotate, for now, and that the satellite is in a slightly elliptical orbit. Then, we have the below diagram:



Similar to before, the light "inherits" the satellite's velocity vector, causing a slight aberration. The path that light appears to take (and thus, what the satellite's radius vector *appears* to be) is $\vec{c} + \vec{v}_{\mathbf{sat}}$

2.3 Realistic Case: Rotating Earth

Let's now consider the fact that the Earth rotates. The observatory's velocity is given by $\vec{v}_{\mathbf{obs}}$. But as a result of the Earth's rotation, the satellite appears to move at backwards, at $-\vec{v}_{\mathbf{obs}}$. It's the same as the slightly harder case, but now with extra satellite movement due to rotation of the Earth



Light appears to follow the path described by $\vec{v}_{\mathbf{apparent}} = \vec{c} + \vec{v}_{\mathbf{sat}} - \vec{v}_{\mathbf{obs}}$
$\vec{v}_{\mathbf{obs}}$ comes from the fact that the Earth is rotating
We also define the following:

- $\vec{c}$, the unaberrated path, is computed by $c\left(\frac{\vec{r}_{\mathbf{sat}} - \vec{r}_{\mathbf{obs}}}{|\vec{r}_{\mathbf{sat}} - \vec{r}_{\mathbf{obs}}|}\right)$, where c is the speed of light (make sure units are consistent)
  - We need a vector with length c
- $\vec{v}_{\mathbf{obs}}$ is the x, y, and z velocities of the observatory
- $\vec{v}_{\mathbf{sat}}$ is the x, y, and z velocities of the satellite
- $\vec{r}_{\mathbf{obs}}$ is the x, y, and z position of the observatory
- $\vec{r}_{\mathbf{sat}}$ is the x, y, and z position of the satellite
- $\vec{v}_{\mathbf{apparent}}$ is the velocity path that light appears to take

We can also compute:

- $\vec{r}_{\mathbf{obs,\ sat}} = \frac{-\vec{v}_{\mathbf{apparent}}}{c}|\vec{r}_{\mathbf{sat}} - \vec{r}_{\mathbf{obs}}|$
  - This tells you the apparent radius vector between the observatory and the satellite
- $\vec{r}_{\mathbf{sat,\ apparent}} = \vec{r}_{\mathbf{obs}} + \vec{r}_{\mathbf{obs,\ sat}}$
  - Tells you the apparent x, y, z position of the satellite

$\vec{r}_{\textbf{sat, apparent}}$ is the most important result. It tells you, "having added in aberration effects, where does the satellite *appear* to be in terms of x, y, and z"

If we know $\vec{r}_{\textbf{obs}}$ and $\vec{r}_{\textbf{sat, apparent}}$, we can do some spherical trig to calculate the altitude and azimuth.

For the purposes of Snell's law, I will define the altitude to be measured from the zenith. An altitude of $0°$ *means straight up.*

I will denote altitude by θ and azimuth by ϕ. Note that neither of these have been affected by atmospheric effects just quite yet.

3 Refraction

As the light from the satellite enters the Earth's atmosphere, the altitude angle gets refracted.

Properly calculating the refraction is nightmarishly hard if we assume a spherical Earth

But, let's examine the index of refraction as a function of height and see if we can make any reasonable approximations.

Source [3] finds that the refractivity, N, can be empirically approximated by:

$$N = 542.9 + 4.854h - 100.86\sqrt{h + 3.919}$$

where h is the height in thousands of feet. Above 101.7464 thousand feet in altitude, $N = 0$

We relate refractivity to refractive index by:

$$n = 1 + \frac{N}{10^6}$$

Essentially, given an observatory height, we can figure out the refractive index it experiences. Now, observe that 102 thousand feet in altitude is only about 19 miles. Since the light does all of its bending really close to the surface, we can assume the curvature of the Earth to be negligible in our refraction estimates.

Applying Snell's law, we find that:

$$\theta_{\textbf{refracted}} = \sin^{-1}\left(\frac{\sin\theta_{\textbf{initial}}}{n}\right)$$

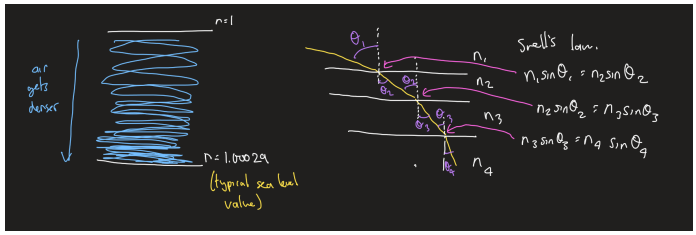
The azimuth angle is unaffected.

At this point, we've added aberration and refraction effects into our observation

Please reference the other document on atmospheric seeing to properly add atmospheric turbulence noise

3.1 Propagated Snell's Law

If you are curious about how we propagated Snell's law, take a look at this diagram:



We can approximate a gradient where the index of refraction changes as a function of height

Consider this:

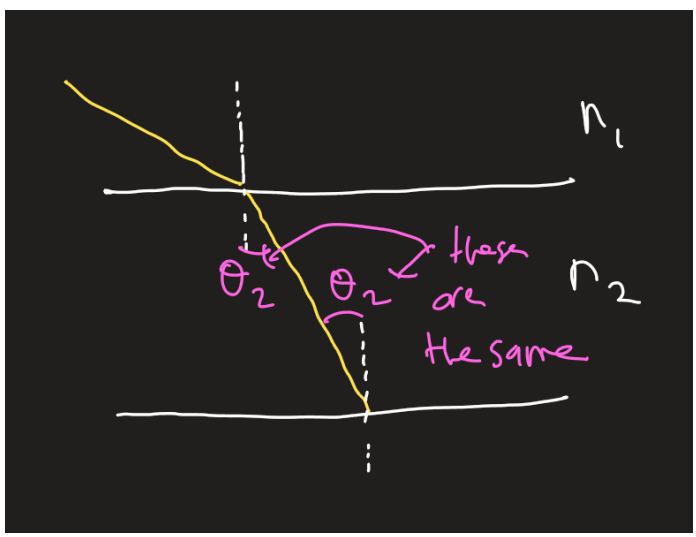
Let's say you have a gradient starting at $n = 1$ (like in space) and ending at $n = 1.00029$ (a typical sea level value)

You can approximate this gradient by subdividing into m intervals, each of which Δh in height

At the interface between the first and second interval, you have

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

Within that second interval, the opposite angle is also θ_2 (see the below diagram)



So, at the interface between the second and third interval, you have:

$$n_2 \sin\theta_2 = n_3 \sin\theta_3$$

By the transitive property, we have:

$$n_1 \sin\theta_1 = n_2 \sin\theta_2 = n_3 \sin\theta_3 = ... = n_m \sin\theta_m$$

Now, let $\Delta h \to 0$ and $m \to \infty$ until we perfectly recreate the original gradient. At each of the interfaces between the infinitesimally thin intervals, Snell's law still applies

4 References

[1] The Things You Can't Ignore: Evolving a Sub-Arcsecond Star Tracker, by John Enright, Doug Sinclair, and Tom Dzamba

[2] LIGHT ABERRATION EFFECT IN HR GEOMETRIC MODEL by D. Greslou, F. De Lussy, J. Montel

[3] A study of radar elevation-angle errors due to atmospheric refraction by M. Fannin and K. H. Jehn